[BP1520] STID Documentation

Content

1. Introduction
2. Updates
3. Scripts Download & File Structures
- Scripts Download
- File Structures

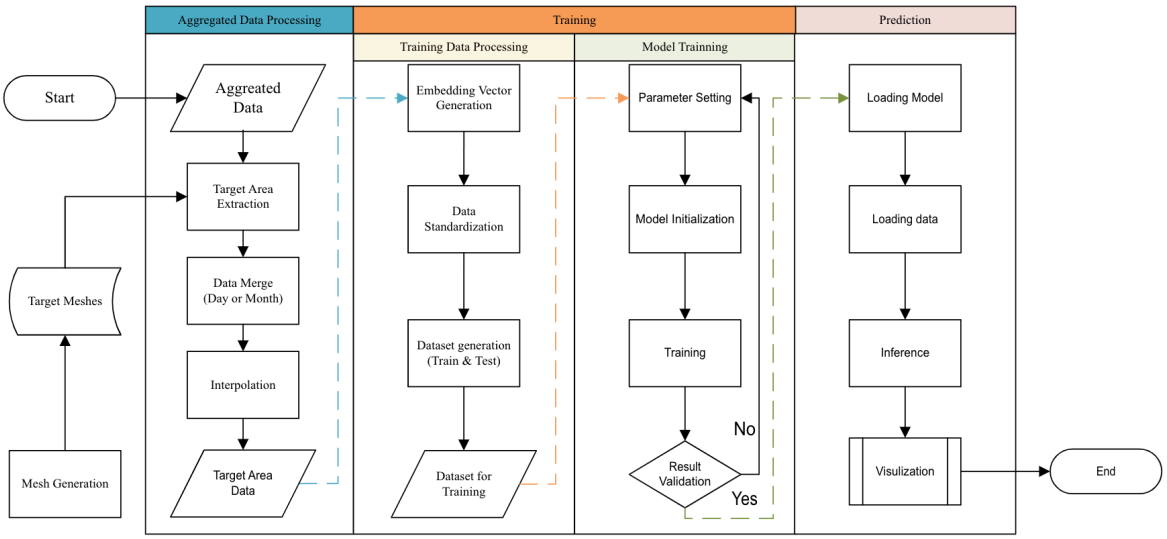
4. Environments
5. Scripts
- run.py
  - Function
  - Parameters
  - Examples
- rawDataProcess.py
  - Function
  - Parameters
- traintest.py
  - Function
  - Parameters
- inference.py
  - Function
  - Parameters

6. Input/Output Data Format
7. Q&A

1. Introduction

This is a documentation for grid-based people flow prediction of project BP1520. Based on the open source model STID (**S**patial **T**emporal **ID**entity) Model. Here is a guide on how to use the provided python scripts to automatically process raw data and train STID models, also conduct inference.

The entire working flow of the provided scripts follows.



Notice that the visualization function is not included in this documentation.

2. Updates

Date	Version	Editor	Summary
2024/10/29	1.0.0-Alpha.1	Mo Yu	Basic data processing, training, inference function implementation.

3. Scripts Download & File Structures

Scripts Download

You can download the zip file here. If you are not authorized to download, please contact `Mo Yu`.

File Structures

If you use linux, you can use the following commands to unzip the file.

```
None
tar -xzvf BP1520_alpha1.tar.gz
```

And the file structure will be like the following screenshot.

/ BP1520 / alpha /

Name	Last Modified
cache	5 hours ago
data	14 days ago
logs	5 hours ago
results	5 hours ago
saved_models	5 hours ago
dataProcess.py	5 days ago
inference.py	51 minutes ago
rawDataProcess.py	32 minutes ago
requirements.txt	6 hours ago
run.py	36 minutes ago
start.json	32 minutes ago
STID.py	14 days ago
traintest.py	2 days ago
utils.py	3 days ago

[**IMPORTANT**] DO NOT MODIFY THE FILE STRUCTURE.
Otherwise, the scripts may fail to work.

4. Environments

To run the STID model successfully, you need to install the necessary libraries. You can use the following command and `requirements.txt` in the zip file to get a quick start.

[Linux]

```
None
python3 -m venv [Your Envs' Path]

source [Your Envs' Path]/bin/activate

cd [Project Folder]

pip install -r requirements.txt
```

5. Scripts

run.py

— main script integrating **data process**, **training** and **inference.**

Function

This script works as the top interface to run STID models, which can be called easily with modifying the parameters in the json file. There are 3 major functions, **raw data processing**, **model training** and **inference(prediction)**. What's more, the script also can help you get the MESH_MAP by specifying the boundary box.

Parameters

Parameters	Type	Default	Description
-p –parameters	str	start.json	Initial parameters (json file) to run the scripts.
-m –mode	int	1	1→ processing & training; 2 → processing 3 → training; 4 → inference
-g –genMesh	bool	False	**True** means generating meshes from specified "bbox". And MESH_MAP will be saved under the root directory of the dataset, like "/data/oneDayData/{City_Name}"
-t –test_data	str	None	File path of test data. If specified, then the specified data will be used.
-d –dataset	str	None	City/Prefecture name of specified "bbox".
-b –bbox	str	None	Bounding box in the format 'left, bottom, right, up'. **Necessary** if -g is True.

PS: The default "start.json" file is like this, it contains all the necessary parameters.


```
▾ root
    dataset "nagoya"
    year "2024"
    month "03"
    input_path "/data/xpop_mesh4_agg"
    output_path "./data/oneDayData/"
    stid_path "./data/trainData"
    data_type "m"
  ▾ training_parameters
      step 12
      batch_size 32
      null 30
      gpu -1
      loss "MAE"
      coeffcient 0.01
      epochs 200
  ▾ inference_parameters
      model_path "./saved_models/STID_nagoya_M_202403_241029234442"
      mesh_file "./data/oneDayData/kochi/meshes.csv"
      data_path "./data/testData/nagoya_l_202403.csv"
      output_path "./results"
```

Examples

Run training process with default parameters.

```
None
mo_yu@lm-saas-sandbox-sipg4:~/BP1520/alpha$ python run.py -m 3 -p start.json
Model Initialization...
Loading dataset located in nagoya_M_202403 ...
No processed data is founded, now processing the raw data...
x shape:  (8905, 12, 4579, 3) , y shape:  (8905, 12, 4579, 1)
nagoya_M_202403 loading is finished!
STID initialization is finished.
Early stopping
Tranning is over, test starts...
Testing is over!
```

Conduct inference with default parameters.

```
None
mo_yu@lm-saas-sandbox-sipg4:~/BP1520/alpha$ python run.py -m 4
Model Loading is finished!
Data is ready, start to inference...
Inferencing finished, ready to write to the output csv file.
Writting is finished! Results are saved in ./results.
```

rawDataProcess.py

Function

This script can help extract data in the target area from the given aggregated data (one day data) and merge the one day data into one month data.

Parameters

Parameters	Type	Default	Description
-d –dataset	str	unknown	Dataset name, important to distinguish
-i –input_path	str	/data/xpop_mesh4_aggs	the input root path of raw data
-o –output_path	str	./data/oneDayData	the output root path of daily aggregated mydaiz data.
-s –stid_path	str	./data/trainData	the output root path of monthly aggregated mydaiz data.
-t –data_type	str	m	Data type of mydaiz data, m-->move, l-->latest.
-y –year	str	2024	The year of input data.
-m –month	str	03	The month of input data.

traintest.py

Function

This script helps train the STID model with training data, and save the best model during the training process which can be used in the after inference part.

Parameters

Parameters	Type	Default	Description
–model	str	STID	Model name to save output in file
-s –step	int	12	Steps to predict, one step = 5 minutes
-b –batch_size	int	32	Number of batches to train in each epoch. Basically, batch_size should be power of 2.
–data_path	str	None	Path of monthly aggregated mydaiz data.
-d –dataset	str	KOCHI	Dataset name which is important to save temporary results and model parameters.
–null	float	30.0	Null value (small) to ignore during training
–loss	str	MAE	MAE, MSE, RMSE, MAPE, Hyrbrid
–coefficient	float	0.01	HYBRID coefficient, e.g. c * MAE + MAPE.
–nodes	int	None	Number of nodes of input data.
–json	str	./start.json	json file contains necessary parameters.

inference.py

Function

Conduct inference/prediction based on selected model.

Parameters

Parameters	Type	Default	Description
–mesh_file	str	./data/oneDayData/kochi/meshes.csv	File path of the mesh mapping file.
-m –mode_path	str	None	Must be specified. Path of the directory saved the model parameters.
-d –data_data	str	./data/test.csv	Path of test dataset file
-o –output_path	str	./results	City/Prefecture name of specified “bbox”.

6. Input/Output Data Format

Basically, the input and output data has the same data format (csv file).

	time	mesh	ex_volume
1	2024–03–02 00:00:00	503301364	104
2	2024–03–02 00:00:00	503301371	132
3	2024–03–02 00:00:00	503301424	66
4	2024–03–02 00:00:00	503301431	100
5	2024–03–02 00:00:00	503301432	77
6	2024–03–02 00:00:00	503301433	78
7	2024–03–02 00:00:00	503301434	102
8	2024–03–02 00:00:00	503301442	83
9	2024–03–02 00:00:00	503301443	127
10	2024–03–02 00:00:00	503301444	317
11	2024–03–02 00:00:00	503301451	106
12	2024–03–02 00:00:00	503301452	202
13	2024–03–02 00:00:00	503301453	101
14	2024–03–02 00:00:00	503301464	76
15	2024–03–02 00:00:00	503301471	99
16	2024–03–02 00:00:00	503301472	78
17	2024–03–02 00:00:00	503301481	112
18	2024–03–02 00:00:00	503301483	86
19	2024–03–02 00:00:00	503301484	106
20	2024–03–02 00:00:00	503301491	855
21	2024–03–02 00:00:00	503301492	247
22	2024–03–02 00:00:00	503301493	90
23	2024–03–02 00:00:00	503301494	81
24	2024–03–02 00:00:00	503301524	79
25	2024–03–02 00:00:00	503301531	100

Sample Data

7. Remarks

- Now, it will be **OOM** to train in Shizuoka prefecture with one month data. (Alpha-1.0.0)
    - At least 64 GB memory is required to load one month's data in Shizuoka.
- When the number of meshes are different in the training set and test set, only the mesh appeared in the training set will have prediction value, others will be set to 0.

8. Q&A